Tectonic Results from Paleomagnetism of the Aphebian Nipissing Diabase at Gowganda, Ontario

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A total of 30 sites (150 cores, 300 specimens) of the 2165 ± ~25 m.y. Nipissing diabase were collected from around the Miller Lake basin at Gowganda. After alternating field demagnetization a stable primary remanence was isolated at 10 sites of which eight were reversely polarized. A fold test shows that the basinal structure is primary at the 95% confidence level. The pole position of 106°W, 14°S ($P_{0.05}$:9°, 14°) is statistically indistinguishable from previous Nipissing poles from the Cobalt and Elliot Lake areas. The evident conclusion is that the Superior and Southern geologic provinces have not undergone significant relative tectonic translation or rotation in the last 2165 m.y. Combining the remanence directions from the 37 available reliable sites of Nipissing diabase, they give a highly reliable pole position of 94.1°W, 14.5°S ($P_{0.05}$:5.7°, 8.8°) for Nipissing time on the polar wander curve for these two provinces of the Canadian Precambrian Shield. Only four of the 37 sites are normally polarized suggesting that the Nipissing event reflects a dominantly reversed paleofield.

Un total de 30 sites (150 carottes, 300 échantillons) du diabase Nipissing de 2165 ± ~ 25 m.a. furent collectionnés autour du bassin de Miller Lake à Gowganda. Après une démagnétisation de champ alternatif une rémanence primaire stable était isolée à 10 sites, desquels huit étaient polarisés en sens inverse. Un test de plissement montre que la structure du bassin est primaire à un niveau de confiance de 95%. La position du pôle de 106°W, 14°S ($P_{0.05}$:9°, 14°) est statistiquement indistincte des pôles antérieurs du Nipissing des régions de Cobalt et Elliot Lake. La conclusion évidente est que les provinces géologiques Supérieur et du Sud (Southern) n'ont pas subi de translation ou rotation tectonique importante durant les derniers 2165 m.a. Combinant les directions de rémanence, à partir des 37 sites disponibles et sûrs du diabase Nipissing, elles donnent une position fort exacte du pôle à 94.1°W, 14.5°S ($P_{0.05}$:5.7°, 8.8°) pour le temps du Nipissing sur la courbe de déplacement polaire pour ces deux provinces du Bouclier Canadien Précambrien. Seulement quatre des 37 sites sont polarisés normalement suggérant que l'événement Nipissing reflète la prédominance d'un paléochamp inversé. [Traduit par le journal]

Introduction

The town of Gowganda is about 190 km north of Sudbury, Ontario (Fig. 1). From 1908 to 1972, native silver was mined near the town from narrow veins in the undulating Nipissing diabase sill. This mining activity has resulted in the sill being well mapped geologically and dated radiometrically. Elsewhere the Nipissing diabase gives good paleomagnetic results. This paleomagnetic study was made to test whether the Miller Lake basinal structure defined by the sill at Gowganda is a primary intrusive or a secondary deformational feature by using the fold test on the sill's remanence directions from sites around the basin (Fig. 2, sites 1–23). This study was also made to test for possible geotectonic movements within the enclosing Superior Geologic Province or between it and other Precambrian provinces by comparing the sill's paleomagnetic pole position to other coeval poles from North America.

Geology

The Gowganda area has been mapped geologically by Collins (1913), Burrows (1921, 1926) and Moore (1955).

The Nipissing diabase intrudes Archean and Huronian rocks (Fig. 2). The Archean rocks include the Keewatin basic to acidic volcanic sequence and a few younger Haileyburian serpentine plugs. These units were severely deformed during emplacement of the Algoman granitic stocks and batholiths. A few Matachewan porphyritic quartz diabase dikes were then intruded. After orogenic deformation, the early Aphebian (Huronian) Cobalt series sediments were deposited. The lower part of the series consists of the conglomerates and greywackes of the Gowganda formation. This formation is disconformably overlain by the feldspathic quartzites and arkoses of the Lorrain formation. The series is gently deformed with dips rarely in excess of 25°. Fairbairn *et al.* (1969) dated the

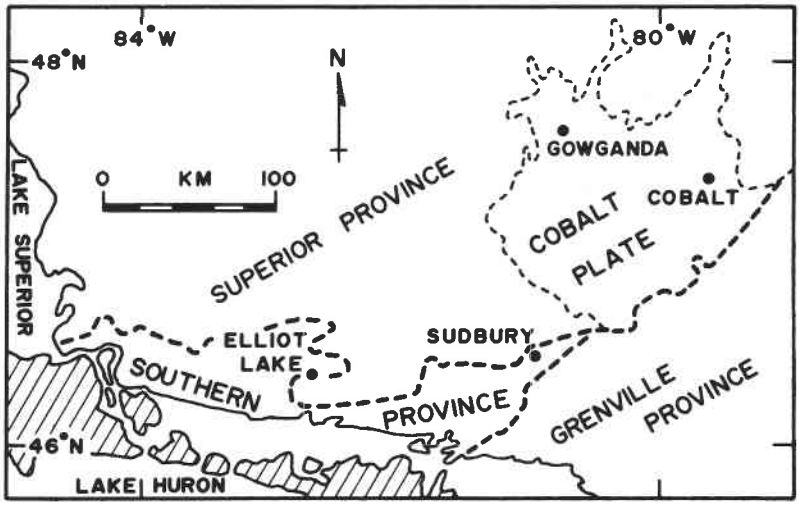


FIG. 1. Location map with the geologic structural provinces and Cobalt Plate subprovince after Stockwell (1969).

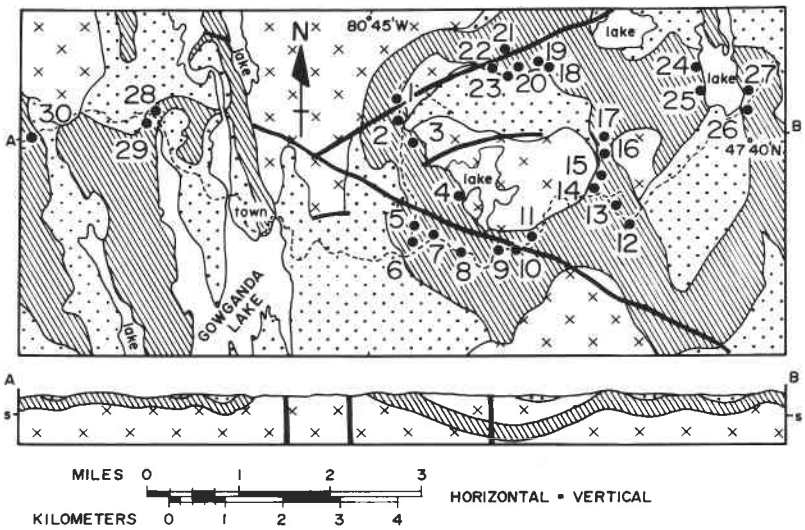


FIG. 2. General geology of the Gowganda Lake area simplified after Moore (1955) with the sampling site locations. Archean units are shown by crosses, Huronian sediments by stippling, Nipissing diabase by diagonal hatching, and post-Nipissing dikes by solid black. The same scale is used in plan and in section A–B with S at sea level.

Gowganda sediments at 2288 ± 87 m.y. using the Rb–Sr whole rock isochron method on 11 samples from near Gowganda.

Nipissing diabase plutons outcrop extensively throughout the Cobalt Plate in the Superior Province and throughout the Penokean Fold Belt of the adjacent Southern Province (Stockwell 1969). The plutons are dominantly conformable sill-like bodies following structural trends in the Huronian sediments with the competent conglomeratic units often providing a barrier to the ascending magma (Card and Pattison 1973). These tabular, sill-like plutons commonly outcrop in elliptical basins and arches which may be entirely primary in origin as sills following folded strata (Miller 1910) or as cone-sheets (Lovell and Caine 1970). Alternatively, the basins and arches may be partly primary and partly secondary (Knight 1922; Thompson 1957). The plutons are also found in transgres-

sive dike-like bodies that apparently follow pre-existing faults.

At Gowganda the Nipissing sill is between 20 and 190 m thick (Moore 1955). The sill dips gently ($\leqslant 25°$) with occasional steep to vertical dips. It is composed of quartz diabase with a feldspar to pyroxene ratio of 1:3 at the bottom varying to 3:1 at the top of the sill. It contains no olivine and only minor biotite. Its ubiquitous titanomagnetite and ilmenite content varies between 1 and 3%. In a few places the quartz diabase has differentiated to granophyre (*e.g.*, sites 14 and 15). The granophyre contains some monoclinic pyroxene and micrographic intergrowths of its quartz and feldspar. It is reddish due to hematite stain and its titanomagnetite content may increase to 5%. The sill's petrology and ferromagnetic mineralogy is about typical of the Nipissing as a whole (Card and Pattison 1973).

Fairbairn *et al.* (1969) dated the Nipissing diabase at Gowganda at 2162 ± 27 m.y. using the Rb–Sr whole rock isochron method on 12 samples from one 270 m core through the sill. This date agrees well with the 2155 ± 80 m.y. Rb–Sr whole rock isochron determined by Van Schmus (1965) for the Nipissing diabase sheets at Blind River (Fig. 1). It also agrees with the Rb–Sr whole rock date of 2180 m.y. (Van Schmus 1965) and the K–Ar biotite date of 2095 m.y. (Lowden *et al.* 1963) determined for the Nipissing diabase sill at Cobalt.

The Nipissing diabase was apparently emplaced in the late stages of the Penokean orogeny some 2165 ± 25 m.y. ago (Church 1968). In adjacent areas to the south around Sudbury, the diabase has been regionally metamorphosed with updated radiometric ages. Thus, some deformation during the Hudsonian orogeny some 1700 m.y. ago is also a possibility.

At Gowganda only a few quartz diabase dikes and one olivine diabase dike were emplaced after the Nipissing diabase and before deposition of the Pleistocene and Recent glacial drift and alluvium. This paucity of younger rocks has prevented determination of the amount of post-emplacement deformation in the Nipissing diabase sill.

Experimental Procedure

Five cores were drilled from each of 30 sites and were oriented *in situ* using a sun compass and/or Brunton compass with topographic checks. Two 2.5 cm right cylindrical specimens were cut from each core. The natural remanent magnetization (NRM) of all 300 specimens were measured on a Schonstedt SSM-1A magnetometer. Using a Schonstedt AF demagnetizer, one pilot specimen per site with an average NRM direction and intensity was step-demagnetized in peak alternating fields (AF) of 50, 100, 150, 225, 350, 500, 750 and 1000 Oe. For each pilot specimen the Paleomagnetic Stability Index (PSI) was calculated and plotted as in Fig. 3 (Symons and Stupavsky 1974). This index measures the rate of change of the remanence direction in millidegrees per oersted (mdeg/Oe) at each AF demagnetizing step. In general, the lowest PSI value indicates the most stable endpoint direction and the optimum AF intensity (H_{opt}) for cleaning or removing unstable remanence components. For each pilot specimen, the ratio of the remanence intensity after each step (J_n) to the original remanence (J_o) was plotted (*e.g.*, Fig. 4). For a stable remanence component this ratio gives a smooth and minimal exponential rate of decay. Based on the results for its pilot specimen, the remaining specimens from the site were AF demagnetized at a selected AF intensity and their remanence remeasured (Table 1). In 26 cores, the remanence directions of the two specimens diverged by more than 20°, thereby indicating inhomogeneous or unstable magnetization, and so the data from these cores were rejected from further consideration. The mean direction was calculated for the remaining cores. In four sites, one core was rejected as anomalous because its remanence direction deviated by more than two times the radius of the cone of 95% confidence (α_{95}; Fisher 1953) of the remaining cores from their mean direction. Using the core mean remanence directions, the site mean directions were calculated along with their α_{95} values (Table 1).

Patel and Palmer (1974) found that the stable remanence in the Nipissing diabase near Elliot Lake resides in magnetite–ilmenite exsolution and that unstable remanence resides in unexsolved titanomagnetite. This explanation is consistent for the mixture of stable and unstable sites found in this study.

Site Mean Directions

The site mean remanence directions were first analyzed at a relatively high level of reliability by accepting only those directions for which the

TABLE 1. Remanent magnetization data by sites

Site	Alternating field step demagnetizing test			Number of cores				Cleaned remanence direction			
	PSI (mdeg/Oe)	H_{opt} (Oe)	H_{mdf} (Oe)	Taken	Rejected (i.m.)	Rejected (anom.)	Used	Length vector R	Declination (degrees)	Inclination (degrees)	Radius of α_{95} (degrees)
1	6.6	750*	35	5	1		4	3.993	29	−47	4.4
2	19.6	750*	65	5	2		3	2.990	18	−43	8.6
3	24.2	500	30	5	3		2	1.992	6	−30	22.9
4	73	50	120	5	2		3	2.399	40	−41	82.5
5	14.3	750	35	5	1		5	3.996	23	−38	9.8
6	8.5	750	45	5			5	4.981	19	−41	5.3
7	6.0	750	730	5			5	4.956	23	−45	8.1
8	2.6	1000	~1500	5			5	4.945	11	−50	8.9
9	20.9	750*	510	5			5	4.915	12	−44	11.3
10	14.2	200*	50	5		1	4	3.942	306	−18	12.9
11	6.0	750	90	5			5	2.711	331	−15	86.6
12	16.7	350	120	5			5	4.905	341	78	11.9
13	14.0	500*	305	5			5	4.950	321	75	8.6
14	53	225	55	5	1		4	3.261	4	51	52.3
15	5.4	350	350	5	1	1	3	2.663	315	68	55.9
16	8.3	500	165	5	1		4	0.166	52	36	>90.0
17	23.3	1000	105	5	3		2	1.983	15	−31	33.7
18	7.7	100*	160	5			5	4.926	1	63	10.5
19	5.0	500	230	5	2		3	2.992	344	78	7.6
20	16.1	350	1050	5	3		2	1.617	334	8	>90.0
21	6.3	500	255	5	3		2	1.995	1	72	18.1
22	22.2	50	180	5			5	4.913	356	75	11.4
23	3.5	350	340	5	1		4	3.315	334	42	49.8
24	55	100	100	5			5	4.966	323	78	7.1
25	2.3	350	135	5			5	4.914	285	70	11.3
26	14.1	1000	50	5	3		2	1.989	349	54	26.3
27	152	100	45	5		1	4	3.921	332	60	16.0
28	47	100	230	5	2		3	2.497	177	68	72.5
29	30	500	105	5	1	1	3	2.933	258	25	22.9
30	30	350	90	5	2		3	2.946	230	39	20.5

NOTES: PSI is the paleomagnetic stability index and H_{opt} is the optimum AF demagnetizing field of Symons and Stupavsky (1974) with the asterisk indicating that $H_{opt} = 1000$ but that a lower field was actually used. Rejected cores are indicated by i.m. for inhomogeneously magnetized with specimen directions diverging by more than 20°, and by anom. for a single anomalously directed core at a site.

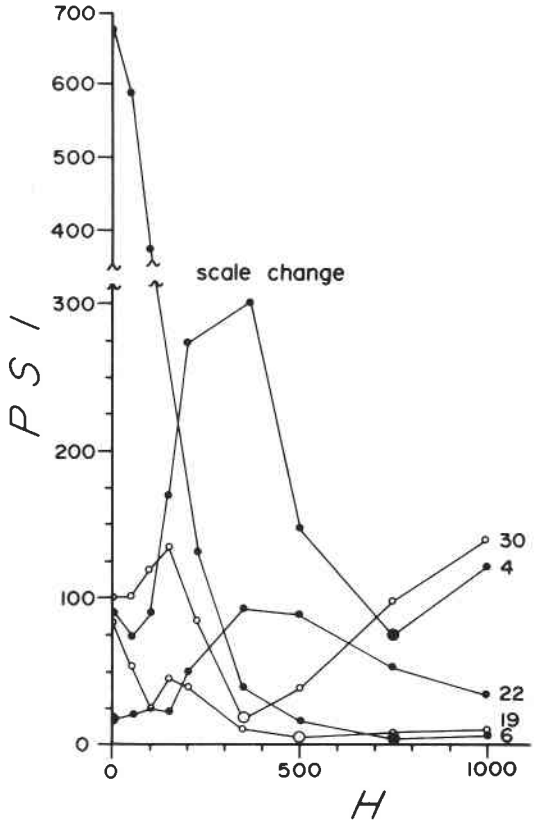


FIG. 3. Typical Paleomagnetic Stability Index curves for specimens from stable (6) and a moderately stable (30) 'reversed group' sites, from stable (19) and moderately stable (22) 'normal group' sites, and from an 'unstable group' (4) site. The PSI is in millidegrees per oersted and it is plotted against the peak demagnetizing field in oersteds with the optimum demagnetizing field (H_{opt}) shown by a larger circle for each specimen.

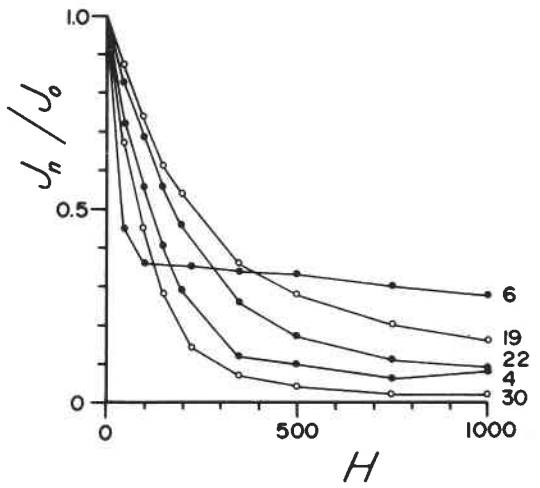


FIG. 4. Typical AF demagnetizing curves shown by plotting the ratio of the remanence intensity at the nth cleaning field to the initial NRM remanence intensity (*i.e.* J_n/J_o) against peak demagnetizing field in oersteds. The same specimens are shown as listed in the caption for Fig. 3.

site $\alpha_{95} \leqslant 10°$. Of the nine accepted directions, six form a reversely magnetized cluster and three a normal cluster. The mean directions and statistics for the reversed and normal groups are shown in Fig. 5a and given in Table 2. The reversed group is thought to record the stable primary direction because it contains the majority of the stable sites and because the pole position calculated from its group mean direction is consistent with other Nipissing poles (Fig. 6). The normal group is thought to record less stable secondary viscous remanent magnetization (VRM) directions acquired during Recent times, mainly because its pole position is coincident with the Earth's present magnetic pole. The same two groups are defined when the level of reliability is decreased to accept the mean directions of those sites with $\alpha_{95} \leqslant 25°$ (Fig. 5b, Table 2). The reverse group now includes sites 29 and 30 with their mean directions reported to the antiparallel position. Also site 10 has an intermediate direction and is excluded from both normal and reverse groups. For the reverse group, inclusion of the less stable sites predictably causes the α_{95} value to increase from 5.0° to 11.3°, the more sensitive precision parameter k of Fisher (1953) to decrease from 184 to 19, and the pole position to change insignificantly by a few degrees. Similarly for the normal group α_{95} goes from 4.9° to 6.4°, k from 625 to 66, and no significant polar shift.

The attitude of the sill was determined for the primary reversed group of sites from adjacent contact measurements or from mine maps. These attitudes were used in a fold test (Graham 1949) to tilt correct the site mean remanence directions by rotating the sill to the horizontal about the strike axis through the dip angle. The tilt corrections increase the scatter in the directions with α_{95} increasing from 11.3° to 19.1° and k decreasing from 19.1 to 7.3 for sites having $\alpha_{95} \leqslant 25°$. Using the comparison of precisions test of Watson (1956), the increase in scatter of the mean directions is significant at the 95% confidence level ($P_{0.05}$). This test indicates that the Nipissing diabase solidified in its present basinal configuration and that it has not been signifi-

TABLE 2. Remanent magnetization data by group of sites

Group of sites	Number of sites	Length of vector resultant R	Mean remanence direction: Declination (degrees)	Inclination + down (degrees)	Precision parameter k	Radius $P_{0.05}$ α_{95} (degrees)	Pole position: Longitude West (degrees)	Latitude North (degrees)	Semi-axes Oval of $P_{0.05}$: δ_p (degrees)	δ_m (degrees)
Sites with $\alpha_{95} \leq 10°$										
Reversed, uncorrected	6	5.9729	20.7	−44.2	184	5.0	99.8	−14.1	3.9	6.2
Reversed, tilt corrected	6*	5.9489	3.5	−59.0	98	6.8	83.4	−2.6	7.6	10.3
Normal, uncorrected	3	2.9968	331.6	77.0	625	4.9	111.2	66.9	8.6	9.2
Sites with $\alpha_{95} \leq 25°$										
Reversed, uncorrected	10†	9.5287	27.3	−42.4	19.1	11.3	106.2	−13.8	8.6	13.8
Reversed, tilt corrected	10*†	8.7739	15.5	−50.9	7.3	19.1	94.2	−9.5	17.4	25.8
Normal, uncorrected	9	8.8782	338.8	73.0	66	6.4	122.2	73.4	10.2	11.4

NOTES: *The attitudes of the Nipissing diabase used for tilt corrections are by site (strike, dip): 1 (N10°W, 27°E), 2 (N10°W, 25°E), 3 (N–S, 26°E), 5 (N45°W, 20°N), 6 (N45°W, 18°N), 7 (N45°W, 20°N), 8 (N70°W, 20°N), 9 (E–W, 20°N), 29 (N35°E, 40°E), and 30 (N18°W, 65°N).
†The polarities of sites 29 and 30 have been reversed.

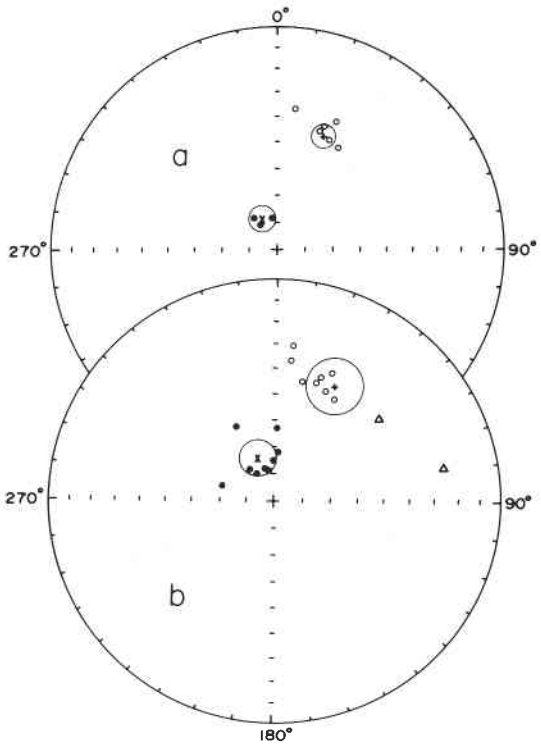


FIG. 5. Site mean remanence directions for sites with (a) $\alpha_{95} < 10°$ and (b) $\alpha_{95} < 25°$. Normal (downward) and reversed (upward) vectors shown by solid and open circles respectively. Sites plotted in their antiparallel position are shown by triangles. The cone of 95% confidence (α_{95} = radius, Fisher 1953) circumscribes the mean position for the primary 'reversed group' (cross) and the secondary 'normal group' (X). E is the present Earth's magnetic field direction.

cantly deformed since emplacement. A similar fold test run on the Nipissing diabase at Cobalt indicated the same conclusion but was not statistically significant (Symons 1970). Thus, the uncorrected remanence directions should be used to calculate pole positions for the Nipissing diabase at Gowganda.

Discussion

There are four previous paleomagnetic studies on Nipissing diabase sills (Table 3). Two studies originate within the Cobalt Plate of the Superior Province. Symons (1967) studied at 65 m vertical section of the sill at the Silverfield Mine near the town of Cobalt (Fig. 1) and found reversed remanence directions only. A more extensive study of the sill around Cobalt revealed a stable reversed remanence in 11 of 14 sites (Symons 1970). The other two studies originate within the Penokean Fold Belt of the Southern Province. Symons (1971) found a stable remanence in two normal and six reversed sites or in eight of 11 sites between Elliot Lake and Matinenda Lake. Patel and Palmer (1974) sampled seven sites to the west of Matinenda Lake and found a stable remanence in five reversely polarized sites.

The 'reverse' group of Nipissing diabase sites with $\alpha_{95} \leqslant 25°$ at Gowganda have been subjected to a similar standard of data screening as used in the previous studies. This group contains eight reversely and two normally polarized site mean remanence directions. This result confirms the earlier observation (Symons 1971) that the Earth's magnetic field was dominantly reversed during emplacement and cooling of the Nipissing sills with minor periods of normal polarity. This observation is true for both the Superior and Southern Provinces, and it also provides substantial evidence that the diabase sills retain a primary remanence of probable TRM origin.

Finally, Symons (1967) used a contact test to show that the diabase sill at the Silverfield Mine retains a primary remanence.

It is visually apparent from the overlapping ovals of 95% confidence in Fig. 6a and from using Watson's (1956) *F*-ratio test that the Gowganda Nipissing population of directions is not significantly different from either the Silverfields Mine or Cobalt area populations. Patel and Palmer (1974) have already noted that the Elliot Lake and Matinenda Lake populations of Nipissing directions are not significantly different at the 95% confidence level. It is further true that the combined Superior Province directions are not significantly different from the combined Southern Province directions (Fig. 6b). The concordance of these populations leads to three conclusions. First, the Nipissing sills in all five collections have retained a stable primary remanence. Second, these sills were emplaced during a single major igneous event. Third, the Superior and Southern Provinces have not undergone significant differential tectonic rotation or translation since emplacement of the Nipissing sills.

The concordance of Nipissing sill directions also allows all the site mean remanence directions to be combined to give a highly reliable overall pole position of 94.1°W, 14.1°S ($P_{0.05}$ = 5.7°, 8.8°) with an assigned age of 2165 $\pm \simeq$25

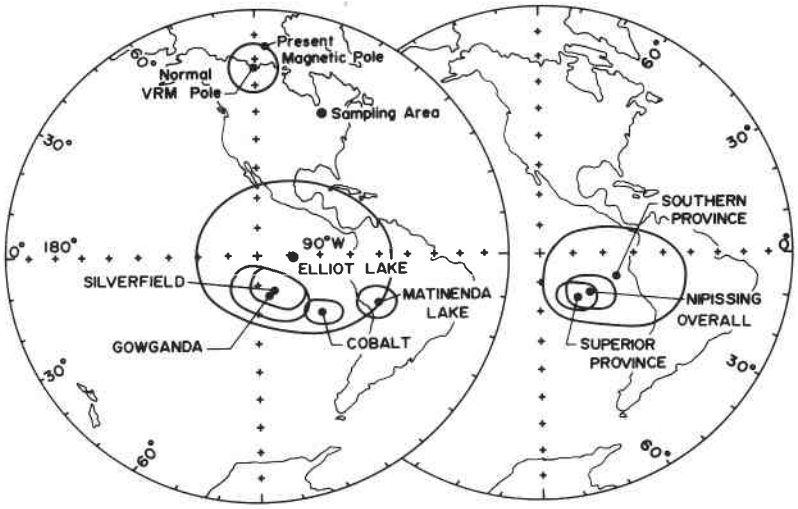


FIG. 6. Pole positions for Nipissing diabase sills as listed in Table 3. The plot is an azimuthal equal-area projection centered on 0°N, 110°W.

TABLE 3. Pole positions for Nipissing sills

Sampling site	Number of sites	Pole position in degrees: Longitude (West)	Latitude (South)	Semi-axes $P_{0.05}$: δ_p	δ_m	Reference
Gowganda area	10	106.2	13.8	8.6	13.8	This study
Cobalt area	11	88.1	19.4	4.0	6.6	Symons (1970)
Silverfield Mine	12*	105.6	11.9	5.7	9.0	Symons (1967)
Superior Province	24*	97.8	16.4	4.3	7.0	
Elliot Lake area	8	98.4	0.8	25.2	33.3	Symons (1971)
Matinenda Lake area	5	69.9	15.6	5.2	7.0	Patel and Palmer (1974)
Southern Province	13	85.1	8.0	16.8	23.6	
Nipissing Overall	37*	94.1	14.1	5.7	8.8	

*The 12 sites at the Silverfield Mine are spaced along a 65 m vertical section of sill, so that this collection has been given a weight of '3 sites' in calculating the Superior Province and Overall mean poles.

m.y. This position is valid for the polar wander curves of the Superior and Southern Provinces and probably of the Slave, Churchill, and Bear Provinces of the North American Precambrian Shield.

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